

YatraSetu

Adaptive, Trust-First AI Tourism Intelligence Platform

Final Product Requirements Document (PRD)

Core product statement

YatraSetu helps travellers decide where to go, whether a destination is right for them now, how to reach it, and what to do when real-world conditions change.

Document item	Value
Problem Statement	SIH 2026 PS ID 26204 - Student Innovation: tourism industry including hotels, travel and others
Theme	Travel & Tourism
Category	Software
Version	1.0 - Final integrated PRD
Product type	AI-powered tourism intelligence + trust + local tourism ecosystem
Primary users	Travellers, local tourism stakeholders, tourism authorities
MVP strategy	Demonstrate the decision engine and an end-to-end adaptive travel journey; phase broader ecosystem modules progressively

Basis of this PRD: the SIH 2026 YatraSetu Solution Guide and the project direction developed in the team discussion. The guide frames YatraSetu around trust, AI personalisation, de-congestion/sustainability, local stakeholder support, offline access, safety, rewards and a tourism-board dashboard; this PRD integrates those foundations with the project's destination-reality, decision, mobility and remote-travel requirements.

Table of Contents

1. Product Overview
2. Problem Statement & Opportunity
3. Product Vision and Principles
4. Goals and Non-Goals
5. Users and Personas
6. End-to-End Product Journey
7. Core Product Architecture
8. Functional Requirements
9. AI & Decision Engine
10. Trust, Verification and Review System
11. Mobility, Vehicle Rental and Arrival Intelligence
12. Safety, Remote Area and Offline Mode
13. Local Tourism Ecosystem
14. Sustainability and De-congestion
15. Stakeholder & Tourism Board Dashboard
16. Data Requirements & Data Quality
17. Technical Architecture
18. Data Model
19. API / Service Requirements
20. Security, Privacy and Responsible AI
21. MVP Scope and Phased Roadmap
22. KPIs and Success Metrics
23. Risks, Dependencies and Mitigations
24. QA and Acceptance Criteria
25. Hackathon Demo Plan
26. Open Decisions and Assumptions
27. Glossary and Reference Baseline

1. Product Overview

YatraSetu is a tourism intelligence platform that adds a decision and adaptation layer on top of fragmented tourism information. The product is not positioned as only a booking engine, map, hotel marketplace or itinerary generator. Its central job is to help a traveller make a better destination and journey decision using personal context plus current destination reality.

The four questions YatraSetu answers

1) Where should I go? 2) Is it right for me right now? 3) How should I reach it? 4) What should I do if reality changes?

1.1 Product pillars

Pillar	Purpose	Primary outputs
Decision Intelligence	Match traveller intent to current destination reality	Suitability, GO/MODIFY/ALTERNATIVE, explanation, confidence
Trust Layer	Increase confidence in stays, operators and reviews	Verified Stay, Verified Review, operator verification, price transparency
Mobility & Resilience	Make travel practical before and during the journey	Route, Smart Arrival Point, vehicle options, mechanic help, remote-area support, offline pack
De-congestion & Sustainability	Distribute tourism demand while preserving intended experiences	Experience-equivalent alternatives, sustainability indicators, demand insights
Local Tourism Ecosystem	Connect travellers with small/local stakeholders	Homestays, guides, vehicle rentals, mechanics, artisans, local experiences

2. Problem Statement & Opportunity

The official SIH problem statement is intentionally broad: a solution/idea that can boost the current situation of the tourism industry including hotels, travel and others. The source solution guide identifies overcrowding, trust deficits, fragmented information, weak support for small stakeholders, language/connectivity barriers and safety concerns as key tourism pain points.

Problem	Observed product gap	YatraSetu response
Reality gap	Online destination information can be static, incomplete or context-free	Destination Reality Engine + source/timestamp/confidence metadata
Personalisation gap	Generic lists do not account for traveller preferences and constraints	Traveller profile + suitability engine
Fragmentation	Weather, mobility, stays, local services and reviews live in separate systems	Unified intelligence layer and normalized data model
Overcrowding	Demand concentrates on well-known destinations	Experience-equivalent alternatives + de-congestion nudges
Trust gap	Reviews/operators/prices may be difficult to verify	Verified Stay/Review + operator verification + price history
Remote travel gap	Support becomes harder with limited connectivity	Remote Area Intelligence + SOS + mechanic + offline mode
Local stakeholder discoverability	Small operators can be harder to surface	Low-friction verified local marketplace

3. Product Vision and Principles

Vision: build a tourism system that can reason about the traveller, destination, mobility and support ecosystem together, instead of merely presenting listings.

3.1 Product principles

- Decision over listing: recommend an action, not just a list of places.
- Reality-aware: dynamic claims should carry source/freshness metadata whenever available.
- Personalised, not universal: a place can be suitable for one traveller and unsuitable for another.
- Experience over coordinates: alternatives should preserve the traveller's intended experience, not merely minimize distance.
- Explainable: users should understand major recommendation factors.
- Trust by design: verification should be a workflow, not a badge added after the fact.
- Safety is contextual: safety indicators show contributing factors and confidence rather than guaranteeing safety.
- Offline resilience: key trip and emergency information must remain available when connectivity is poor.
- Provider-agnostic: the decision engine should not depend on one map, transport or mobility provider.
- Human override: plans remain editable and users retain control over recommendations and bookings.

4. Goals and Non-Goals

4.1 Goals

- Provide a single, understandable decision flow from traveller intent to destination suitability and travel action.
- Combine static and dynamic tourism signals into a normalized destination context.
- Support GO/MODIFY/ALTERNATIVE outcomes with explanations.
- Recommend experience-equivalent alternatives when a chosen destination becomes unsuitable.
- Make mobility practical through routes, Smart Arrival Points and provider listings.
- Support remote-area travel with safety information, SOS, mechanics and offline access.
- Increase trust in local stakeholders, stays and reviews.
- Support sustainable footfall distribution and local economic discovery.
- Provide a feasible MVP that can be demonstrated with a limited pilot region.

4.2 Non-goals for the MVP

- Building a nationwide real-time crowd sensing network.
- Replacing every OTA, airline, railway, bus operator or ride-hailing service.
- Scraping restricted platforms or bypassing third-party API terms.
- Building a full blockchain infrastructure; a tamper-evident hash chain is sufficient for the MVP.
- Guaranteeing safety, weather or crowd conditions with absolute certainty.
- Implementing every tourism stakeholder workflow end-to-end in the first release.

5. Users and Personas

Persona	Needs	Representative jobs-to-be-done
Traveller - planner	Personalised discovery and confidence	"Find a place that fits my budget, interests and time."
Traveller - active trip	Fast adaptation and support	"Conditions changed; tell me what to do next."
Remote-area traveller	Connectivity-independent support	"I am travelling far from cities; keep the essential information available."
Local operator	Discovery and verified presence	"Let legitimate travellers find my service without heavy onboarding."
Tourism authority	Demand visibility	"Understand where tourists are going and where demand can be distributed."

5.1 Primary traveller profile fields

- Interests and intended experiences
- Budget range
- Travel dates / duration
- Number of travellers
- Travel pace
- Crowd preference
- Accessibility requirements
- Transport preference
- Vehicle requirement
- Language preference
- Emergency contact(s)

6. End-to-End Product Journey

The product should be explainable as one journey even though the underlying system contains multiple modules.

Stage	Product behavior
Understand	Capture traveller context and intended experience.
Discover	Find destinations compatible with intent and constraints.
Assess	Evaluate destination reality using available dynamic and static signals.
Decide	Return GO, MODIFY or ALTERNATIVE with reasons and confidence.
Reach	Generate route, choose Smart Arrival Point and surface vehicle/rental options.
Support	Provide offline information, SOS, remote-area assistance and mechanic help where relevant.
Adapt	React to weather, closures, crowd changes or transport disruptions.
Learn	Capture actual experience and compare it with the prediction.

Canonical user journey

Traveller profile → Discovery → Reality Check → Suitability → GO/MODIFY/ALTERNATIVE → Mobility → Support → Adaptive itinerary → Feedback

7. Core Product Architecture

The architecture separates experience, decision intelligence, normalized data and external providers. This allows the product to evolve without coupling the core decision logic to one external platform.

- Client Apps: Traveller mobile/web experience + Stakeholder/Tourism Board dashboard
- API Gateway: authentication, authorization, rate limiting and routing
- Core Services: Profile, Discovery, Reality, Decision, Mobility, Trust, Safety, Marketplace, Offline, Feedback, Analytics
- Decision Layer: scoring/rules + AI services for intent, review analysis, experience similarity and explanations
- Data Layer: PostgreSQL + cache + geospatial indexes + evidence/source metadata + audit/hash log
- External Integrations: weather, maps, public transport, notifications, government/open data and permitted place/review data

8. Functional Requirements

8.1 Traveller onboarding and profile

ID	Requirement	Priority	Acceptance criteria
FR-PROF-01	System shall capture interests, budget, dates, party size and pace.	P0	Profile can be created and edited; missing required fields are validated.
FR-PROF-02	System shall capture crowd and accessibility preferences.	P0	Preferences influence suitability and ranking.
FR-PROF-03	System shall allow users to specify an intended experience in natural language or tags.	P0	Input is normalized into experience attributes.
FR-PROF-04	System shall keep plans editable by the traveller.	P0	User can modify itinerary items without breaking the trip state.

8.2 Destination discovery and reality

ID	Requirement	Priority	Acceptance criteria
FR-DEST-01	Discover destinations based on profile and experience intent.	P0	Results contain fit rationale.
FR-DEST-02	Display static destination facts and facilities.	P0	Location, categories, hours and facilities are shown when available.
FR-DEST-03	Aggregate dynamic signals.	P0	Weather/crowd/operations/recent feedback appear with freshness metadata when available.
FR-DEST-04	Show a confidence indicator for major suitability outputs.	P1	Confidence is distinct from suitability.
FR-DEST-05	Allow user to assess a destination they already selected.	P0	Assessment can be started without re-running discovery.

8.3 Decision engine

ID	Requirement	Priority	Acceptance criteria
FR-DEC-01	Return GO/MODIFY/ALTERNATIVE based on traveller profile and current context.	P0	Decision is deterministic for the same inputs within a versioned rules/model state.
FR-DEC-02	Explain major factors behind the decision.	P0	At least 3 useful reasons or factors are displayed when available.
FR-DEC-03	Recommend experience-equivalent alternatives.	P0	Alternative shares core intended experience attributes and is screened for current conditions.
FR-DEC-04	Do not treat the geographically nearest alternative as the default.	P0	Experience similarity and constraints are part of ranking.

8.4 Mobility and arrival

ID	Requirement	Priority	Acceptance criteria
FR-MOB-01	Generate a route using available transport/mapping data.	P0	Route has origin, destination, legs and estimated time/distance when data exists.
FR-MOB-02	Select a Smart Arrival Point based on operational usefulness, not distance alone.	P0	Ranking considers walking/accessibility + facilities + transport connectivity where available.
FR-MOB-03	Surface vehicle rental options.	P1	Eligible local rental providers can be listed with verification status and contact/booking path.
FR-MOB-04	Support last-mile planning.	P1	Walking/local transport segment can be represented.

8.5 Safety, remote area and offline

ID	Requirement	Priority	Acceptance criteria
FR-SAFE-01	Provide one-tap SOS workflow.	P0	User can trigger SOS; system displays/shares last known location and emergency contacts when available.
FR-SAFE-02	Show a Safety Indicator with contributing factors.	P1	Indicator is accompanied by factor list and freshness/unknown states.
FR-SAFE-03	Detect or flag remote-area conditions using configured thresholds/data.	P1	User receives a remote-area information card when criteria are met.
FR-SAFE-04	Provide mechanic assistance discovery for vehicle breakdowns.	P1	Nearby verified mechanics can be listed; fallback contact path is shown if data is unavailable.
FR-SAFE-05	Create an offline trip pack.	P0	Core itinerary, essential map data, emergency contacts and saved provider info can be viewed without network after download.

8.6 Trust and reviews

ID	Requirement	Priority	Acceptance criteria
FR-TRUST-01	Mark verified stays/operators.	P0	Verification status is visible and tied to a verification record.
FR-TRUST-02	Unlock review submission after verified stay/check-in flow.	P0	Unverified users cannot submit a verified-stay review.
FR-TRUST-03	Maintain tamper-evident review record.	P1	Hash-chain/audit entry exists for verified review events.
FR-TRUST-04	Show available price-history information.	P1	History is time-stamped and distinguishes observed vs user-entered data.

9. AI & Decision Engine

The AI layer should be used where language understanding, semantic similarity and unstructured content processing provide value. Deterministic algorithms should handle routing, distances, timetable matching, geo-filtering and hard constraints.

9.1 AI responsibilities

- Traveller intent extraction from text or structured responses.
- Experience representation, such as history + photography + peaceful environment.
- Review sentiment/topic extraction and complaint clustering.
- Experience-equivalent destination matching.
- Natural-language recommendation explanations.
- Itinerary drafting and adaptation suggestions.
- Multilingual/voice understanding where enabled.

9.2 Non-AI responsibilities

- Distance and route calculation.
- Transport schedule and stop matching.
- Hard business constraints: opening hours, budget ceiling, accessibility constraints, reservation status.
- Geospatial radius and network analysis.
- Offline synchronization and cache management.
- Verification state machine and audit logging.

9.3 Suitability model

For the MVP, use a transparent weighted scoring layer. Exact weights should be configurable and versioned rather than embedded in UI code.

Signal group	Example inputs	Role
Traveller fit	Interest match, budget, pace, accessibility	Hard/soft constraint
Environment	Weather, rain, temperature	Suitability adjustment
Crowd	Current/estimated crowd vs preference	Suitability adjustment
Operations	Open/closed, restrictions, event conflicts	Hard constraint
Mobility	Travel time, connectivity, last-mile difficulty	Suitability/plan quality
Support	Accommodation, medical, toilets, food, mechanic, emergency access	Risk/resilience context
Experience	Semantic similarity to intended experience	Alternative quality

Important safety rule

Suitability score is a planning aid, not a guarantee. Safety and dynamic conditions should display evidence, freshness and unknown states when data is incomplete.

9.4 GO / MODIFY / ALTERNATIVE logic

- GO when the destination satisfies hard constraints and the combined context is compatible with traveller preferences.
- MODIFY when the destination remains viable but timing, route, arrival point or activity can materially improve the outcome.

- ALTERNATIVE when one or more hard constraints fail or the destination is materially incompatible with the traveller context; preserve experience intent through similarity matching.

10. Trust, Verification and Review System

The trust layer is a core product capability, not a visual badge system. The source guide proposes verified stays, KYC-style operator checks and a tamper-evident review history.

10.1 Verification states

Entity	States	Examples of evidence
Stay	Unverified / Verified / Suspended	Booking + check-in confirmation
Guide	Pending / Verified / Suspended	Identity/registration documents and review history
Vehicle rental	Pending / Verified / Suspended	Owner/business verification and vehicle documents
Mechanic	Pending / Verified / Suspended	Provider identity, service area, business/contact evidence

10.2 Review lifecycle

1. Booking or provider interaction creates an eligible experience record.
2. Check-in or other verification event marks the experience as completed/eligible.
3. Review becomes available only after eligibility is confirmed.
4. Review is stored with source, timestamp and verification state.
5. Hash/audit chain records integrity events for tamper evidence.
6. Moderation handles abuse, spam, privacy and prohibited content separately from verification.

11. Mobility, Vehicle Rental and Arrival Intelligence

11.1 Smart Arrival Point

The system should optimize for operational usefulness rather than raw geographic distance. Example factors: walking difficulty, transit connectivity, hotel proximity, toilet access, medical access, parking, food and local service availability.

11.2 Vehicle rental

- Vehicle type: bike, scooter, car or other configured category.
- Trip fit: terrain, distance, party size and budget.
- Provider verification status and availability.
- Price display and rental terms where available.
- Pickup/return location linked to the travel plan.
- Escalation to support/mechanic if a trip involves remote areas.

11.3 Mechanic assistance

- Breakdown flow can be initiated manually or through a trip context shortcut.
- Search nearby mechanics using geospatial data and service capability.
- Show availability, distance, verification state and contact path.
- Support remote fallback contacts when no verified mechanic exists nearby.
- Log the assistance event against the trip only when the user consents.

12. Safety, Remote Area and Offline Mode

12.1 Safety Indicator

The Safety Indicator should be presented as a contextual information layer, not a promise of safety. The system should list the factors used and explicitly surface missing data.

- Connectivity availability/quality where a reliable source exists
- Access to emergency facilities
- Road/route constraints when available
- Distance to assistance providers
- Recent verified user reports
- Remote-area status
- Data freshness and confidence

12.2 Remote Area Intelligence

When a trip enters a configured remote-area context, surface a preparedness checklist: offline pack status, emergency contacts, nearest known medical support, mechanic/fuel availability where available, last known connectivity information, accommodation and transport fallback.

12.3 Offline mode

Offline asset	Required	Behavior
Itinerary	Yes	Day-wise plan and key changes saved locally
Emergency contacts	Yes	Phone numbers and instructions accessible without network
Saved providers	Yes	Hotel, rental, mechanic and guide contact/details from last sync
Essential maps/routes	Yes for configured pilot	Use downloadable/offline-capable map data where licensing permits
Live crowd/weather	No	Display last sync timestamp; do not imply live state offline
SOS	Best effort	Trigger available device/communication mechanisms; show limits when network is unavailable

13. Local Tourism Ecosystem

YatraSetu should support hotels, homestays, local guides, vehicle rentals, mechanics, artisans and local experience providers. The goal is discoverability and trusted connection, not an unlimited marketplace in the MVP.

Stakeholder	Core capabilities	MVP treatment
Hotels / homestays	Listing, verification, availability, booking path	Live or curated listings
Guides	Profile, verification, service area, contact/booking	Curated/verified sample providers
Vehicle rentals	Vehicle type, location, availability, verification	Listing + contact/booking mock
Mechanics	Service area, capability, verification, availability	Nearby provider discovery
Artisans / local experiences	Discovery, category, local benefit	Curated listing; later transaction workflow

14. Sustainability and De-congestion

The de-congestion engine should redirect demand only when the alternative preserves meaningful traveller intent. It should not treat all crowd reduction as universally positive.

- Map popular destinations to experience-similar alternatives.
- Consider crowd pressure and environmental/local context where reliable data exists.
- Prefer better-timed visits when changing the time solves the issue.
- Surface lesser-known destinations as contextual alternatives rather than as a standalone catalogue.
- Expose an interpretable sustainability indicator and its input factors.
- Provide aggregated demand distribution to tourism authorities without exposing personal traveller data.

15. Stakeholder & Tourism Board Dashboard

15.1 Local operator dashboard

- Manage listing information
- Submit/renew verification documents
- Manage availability where supported
- View booking/lead indicators where supported
- Respond to review or verification issues

15.2 Tourism board dashboard

- Anonymised demand distribution
- Destination demand trends
- Popular vs underutilised destination patterns
- Aggregate sustainability/footfall indicators
- Potential intervention areas for infrastructure and visitor management

16. Data Requirements & Data Quality

Data domain	Examples	Preferred sources	Freshness
Destination master	Name, coordinates, type, hours, fees, facilities	Tourism authorities, curated/open data, permitted places APIs	Slow-changing; versioned
Weather	Current conditions, forecast, warnings	Permitted weather APIs	Minutes-hours
Crowd	Observed/estimated crowd	Official feeds where available, permitted signals, user reports	Near-real-time when available
Transport	Routes, stops, schedules	Government/public feeds, permitted providers	Schedules + live when available
Reviews	Text, date, verified status, topics	Permitted review data, first-party reviews	Continuous
Operators	Identity/verification, service area	Operator submission + verification workflow	On change / expiry
Support facilities	Hospital, police, mechanic, fuel, toilet, food	Open geospatial data, verified providers	On change

16.1 Data evidence model

Each dynamic observation should carry: value, source, collected_at, expires_at or freshness window, confidence, evidence type, and optional location scope. Unknown is a valid state.

16.2 Data fallback hierarchy

7. Official government/destination source
8. Verified provider source
9. Multiple independent observations
10. Individual user report
11. Model inference (must be labeled as inference)

17. Technical Architecture

Layer	Recommended components	Responsibility
Client	React / Flutter / React Native	Traveller UI, offline cache, SOS, itinerary, marketplace
API Gateway	Node.js/Express or FastAPI	Auth, rate limiting, service routing
Core services	Profile, Discovery, Reality, Decision, Mobility, Trust, Safety, Marketplace, Feedback	Business logic and orchestration
AI services	LLM + embeddings/similarity + classifier/scoring components	Intent, reviews, experience matching, explanations
Data	PostgreSQL + PostGIS where available + Redis/cache	Profiles, destinations, trips, providers, reviews, geospatial and cached data
Audit	Hash-chained log / managed service later	Tamper evidence for verified review events
Integrations	Maps, weather, transport, notifications, government/open data	External data and actions
Hosting	Vercel/Render/cloud provider / equivalent	Deployment, scaling and operations

18. Data Model

The following entities form the minimum conceptual data model. Exact schema can evolve during implementation.

Entity	Key fields	Relationships
User	id, preferences, accessibility, language, emergency_contacts	Owns trips, feedback, bookings/reviews
TravellerIntent	trip_id, experience_tags, constraints	Input to discovery/decision
Destination	id, geo, categories, hours, facilities	Has observations, alternatives, providers
DestinationObservation	type, value, source, collected_at, confidence	Attached to destination/route
ExperienceProfile	tags, embedding/version, weights	Used for similarity and alternatives
Itinerary	trip_id, days, items, version	Changes over time; offline copy
ItineraryEvent	type, time, reason, old/new value	Auditable plan changes
Provider	type, verification_status, geo, service_area	Hotel/guide/rental/mechanic/artisan
Booking/ServiceRequest	provider_id, trip_id, status	Links user to services
VerificationRecord	entity_id, evidence, status, timestamps	Trust state
Review	user_id, provider/destination, verified_stay, text, status	Eligible only after required verification
AssistanceEvent	type, geo, provider_id, status	SOS/mechanic support
Feedback	prediction_id, actual_experience, delta	Learning loop

19. API / Service Requirements

Endpoint/service	Purpose	Example payload concepts
POST /profile	Create/update traveller profile	preferences, budget, pace, accessibility
POST /discover	Find destination candidates	intent, dates, location, constraints
GET /destinations/:id/reality	Get destination reality context	weather, crowd, operations, evidence
POST /decision/evaluate	Evaluate selected destination	profile + destination context -> decision
POST /alternatives	Find experience-equivalent alternatives	experience vector + constraints
POST /itinerary/generate	Generate/edit itinerary	days, destinations, mobility constraints
POST /itinerary/replan	Adapt itinerary	trigger + current plan + context
GET /mobility/options	Return transport paths	origin, destination, time
GET /arrival-points	Rank operational arrival points	destination + traveller constraints
GET /providers	Find hotels/guides/rentals/mechanics	type + geo + verification
POST /assist/sos	Start emergency workflow	trip_id, last_known_location
POST /assist/mechanic	Request/find mechanic	vehicle_type, geo, issue
GET /offline-pack:trip	Get package metadata/assets	trip version + assets
POST /feedback	Record actual experience	prediction_id + actuals

20. Security, Privacy and Responsible AI

- Encrypt data in transit and at rest using platform-standard controls.
- Use role-based access for traveller, operator and tourism-board dashboards.
- Do not expose precise traveller location to operators or dashboards unless explicitly required and consented.
- Store only the personal data required for the product journey.
- Emergency location sharing should be consent-driven, visible to the user and time-bounded when practical.
- Separate anonymised analytics from personally identifiable traveller data.
- Maintain audit trails for verification and sensitive administrative actions.
- Do not present model inference as verified fact.
- Expose source/freshness/confidence for dynamic recommendations when available.
- Allow manual user correction and itinerary editing.

20.1 Responsible AI requirements

- Model output should never silently override hard safety or operational constraints.
- LLM-generated explanations must be grounded in the data actually used for the recommendation.
- Unknown or stale inputs should lower confidence or trigger an explicit limitation state.
- Recommendation models should be versioned so results can be reproduced or audited.
- Human review remains available for operator verification and safety escalation workflows.

21. MVP Scope and Phased Roadmap

Phase	Scope	Success gate
Phase 0 - Prototype foundation	Curated destinations, profile, weather, static facilities, basic transport/map integration	Data loads reliably and key journey can be demoed
Phase 1 - SIH MVP	AI itinerary, reality assessment, GO/MODIFY/ALTERNATIVE, experience alternatives, Smart Arrival Point, verified review simulation, SOS, offline pack, Yatra Points	Complete end-to-end demo with 1-2 pilot regions
Phase 2 - Pilot	Operator verification, vehicle rental, mechanic discovery, better transport feeds, richer feedback loop, stakeholder dashboard	Real pilot users complete trips and report useful recommendations
Phase 3 - Scale	More regions, multilingual/voice depth, real demand analytics, provider expansion, richer adaptive planning	Stable regional rollout and repeat usage

22. KPIs and Success Metrics

Metric	Definition	MVP target style
Decision usefulness	Users who accept or edit the recommendation positively	Measure baseline; improve through testing
Alternative relevance	Share of alternatives judged to preserve intended experience	Human-evaluated benchmark set
Reality freshness	Share of dynamic observations within their freshness window	Track by source/domain
Explanation coverage	Recommendations with explicit usable reasons	High coverage for P0 flows
Adaptive response success	Trips successfully replanned after a simulated trigger	End-to-end test metric
Offline readiness	Critical trip assets accessible after download without network	100% for defined pack contents
Trust conversion	Eligible stays that produce verified reviews	Track after pilot
Local discovery	Lesser-known destination interactions/visits generated through alternatives	Track after pilot
Support response	Successful mechanic/SOS handoff events	Track with simulated/real logs

23. Risks, Dependencies and Mitigations

Risk / dependency	Impact	Mitigation
Limited API/data availability	High	Use provider abstraction; pilot with curated/open/official data; never rely on scraping.
Crowd data is incomplete	High	Use multi-source evidence, user reports and explicit confidence/unknown states.
Cold-start on local providers	Medium	Start with a small pilot region; low-friction verification and curated onboarding.
Trust in AI recommendation	High	Explain reasons, expose confidence, keep itinerary editable.
Offline map licensing/storage constraints	Medium	Use only legally permitted downloadable data; keep essential non-map data always offline.
Safety expectations exceed actual capability	High	Clearly label limits; SOS is best-effort and does not replace emergency services.
Feature sprawl	High	Prioritize core decision loop; treat marketplace modules as supporting capabilities.
Provider API terms	High	Review terms before integration; use official endpoints/partnerships only.
Sensitive location data	High	Consent, minimization, access control and short retention where possible.

24. QA and Acceptance Criteria

24.1 Core scenario tests

- Traveller wants a peaceful heritage/photography experience; selected hotspot has high crowd -> system suggests an experience-equivalent alternative.

- Destination has bad weather but remains viable later -> MODIFY with time change and explanation.
- Destination closes unexpectedly -> itinerary replans to another suitable activity/destination.
- Selected arrival point is geographically closest but has poor access -> Smart Arrival Point selects a more useful point.
- Road trip enters remote area -> remote-area card appears and offline pack is prompted.
- Vehicle breakdown -> mechanic discovery shows nearby verified provider or fallback path.
- No network after download -> core itinerary, saved providers and emergency information remain accessible.
- Review submitted before eligible check-in -> submission blocked.
- Dynamic data is stale -> UI shows timestamp/unknown state instead of pretending it is live.

24.2 Acceptance gates for the SIH demo

- A new traveller can complete the questionnaire in under two minutes.
- The system produces a day-wise itinerary and identifies at least one destination decision.
- The selected destination displays reality signals and a transparent recommendation.
- At least one alternative preserves intended experience and is not simply the nearest location.
- Smart Arrival Point demonstrates a non-trivial operational choice.
- Offline mode can be demonstrated by disabling network after downloading the trip pack.
- SOS and mechanic flows can be demonstrated in simulation without claiming live emergency integration if not connected.
- Verified review flow prevents an unverified review and allows an eligible review.

25. Hackathon Demo Plan

The demo should tell one coherent story instead of opening many unrelated screens.

Time	Demo beat	What judges should understand
0:00-0:30	Problem	Tourists can find destinations but struggle with current suitability and fragmented reality.
0:30-1:30	Profile + intent	The product understands who the traveller is and what experience they want.
1:30-2:30	Reality check	Current context changes the destination decision.
2:30-3:30	GO/MODIFY/ALTERNATIVE	The system acts as a decision engine, including experience-equivalent alternatives.
3:30-4:15	Mobility	Smart Arrival Point + route + vehicle option.
4:15-5:00	Remote resilience	Offline + SOS + mechanic scenario.
5:00-5:45	Trust	Verified stay/review mechanism.
5:45-6:30	Impact	De-congestion, local stakeholder discovery and tourism-board value.

One sentence to close the pitch

YatraSetu is not another tourism listing app; it is a decision, trust and adaptation layer that helps travellers make better choices as real-world travel conditions change.

26. Open Decisions and Assumptions

12. Select the first pilot geography and define the curated destination set.
13. Select legally usable map/places/review/weather/transport data providers for the pilot.
14. Finalize suitability weights and the minimum hard constraints for GO/MODIFY/ALTERNATIVE.
15. Define the exact verification method for stays, guides, vehicle rentals and mechanics.

16. Choose the offline map/data strategy and storage limits.
17. Decide whether booking is real, deep-linked or simulated for the SIH prototype.
18. Define the emergency escalation path and clarify what is simulated vs truly integrated.
19. Define retention/consent rules for location, SOS and feedback data.
20. Establish the benchmark set for alternative relevance and recommendation quality.

27. Glossary and Reference Baseline

Term	Meaning in YatraSetu
Destination Reality	Current contextual state of a destination derived from available dynamic and static signals.
Suitability	How well a destination fits a specific traveller and current conditions.
GO / MODIFY / ALTERNATIVE	Primary decision outcomes.
Experience-equivalent alternative	Different destination that preserves the traveller's intended experience attributes.
Smart Arrival Point	Operationally useful arrival/drop point selected using more than geographic distance.
Safety Indicator	Contextual safety/support summary with contributing factors; not a guarantee.
Remote Area Intelligence	Contextual support for travel where connectivity and nearby services may be limited.
Offline Pack	Saved itinerary and essential trip/support information available without network.
Verified Stay	Stay associated with a verification event such as booking + check-in.
Tamper-evident review	Review record whose integrity can be detected through an audit/hash chain.
De-congestion	Redirecting or re-timing tourist demand to reduce pressure on over-visited destinations.

Reference baseline: SIH 2026 YatraSetu Solution Guide, PS ID 26204. The source guide describes the tourism problem, trust layer, AI personalisation, de-congestion/sustainability, offline-first access, safety/SOS, Yatra Points, stakeholder dashboard, recommended technology stack, MVP scope, feasibility and expected impact. The present PRD retains that framing and expands it with the project's final decision-engine, destination-reality, mobility, vehicle-rental, mechanic, remote-area and adaptive-travel requirements.